GAUGE, NOT LEVER: DISSOCIATING READABLE CORRECTNESS FROM ITS CAUSAL REPAIR CHANNEL IN A 27B LANGUAGE MODEL

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ABSTRACT

Large language models linearly encode whether they are about to answer correctly (probe AUC up to 0.94), and recent work shows this readout resists steering — but why, and what *does* causally control correctness? On depth-controlled ontology-reasoning tasks with Gemma 3 27B, we first show the readout is not necessary: deleting everything linearly readable about correctness — the full rank-9 INLP subspace at all five readable layers, every token — has no detectable cost ($+0.05$ $[-0.07, +0.20]$; MDE ≈ 0.13), while matched-rank random subspaces are catastrophic (-0.38). Correctness can instead be repaired through a compact channel at concept-mention tokens: a rank-8 basis repairs held-out failures ($+0.25$), passing random-subspace and matched-noise specificity controls. A natural correct-minus-incorrect class-mean vector leaves no detected effect at full dimension yet repairs strongly ($+0.34$; paired projected$-$raw $+0.30$) when concentrated into that channel; shuffled-label and sign-flip controls show the labels carry the effect, and a donor-free variant — answer-free in content, though not yet in addressing — reaches $+0.40$ on a 0.12 baseline where majority-vote self-consistency scores 0.000. The channel is causally potent yet not decodably privileged (probe accuracy within the random-subspace null's upper tail, under its pre-registered threshold) — the inverse of the field's decodable-but-not-causal refrain — and removal experiments show its site demands the state's content wholesale: the lever is write-compact, what it writes into is carried broadly, and the carried content is anchored at the lever's site (matched destruction elsewhere costs a fraction). Positive results are one model and task family, with the raw-axis non-necessity replicating in a second model; every causal experiment was pre-registered before unblinding (31 verdicts, 9 confirmed directional predictions; branch-complete decision rules on all confirmatory items).

1 INTRODUCTION

A linear probe on a language model's residual stream can tell you, before generation begins, whether the answer about to be produced will be correct. This is by now a replicated fact (arXiv:2605.05715, 2026; arXiv:2604.13068, 2026; arXiv:2604.05655, 2026). It is also, on its face, an invitation: if the model internally represents "I am about to get this wrong," surely writing the opposite into the residual stream should help. We spent months accepting that invitation — raw-direction steering, optimized vectors, low-rank interchange over the probe subspace, decode-time gated injection — and repaired nothing. Independently and concurrently, three groups hit the same wall: the failure signal is decodable but steering it is null (arXiv:2605.05715, 2026; arXiv:2604.13068, 2026), a phenomenon one group names *epiphenomenal correctness* (arXiv:2605.23315, 2026). The field's diagnosis stops at the null. This paper asks the question the other way around: if the readable signal is not how correctness is controlled, what is — and what, precisely, is the readable signal *for*?

We work on abductive ontology reasoning (InAbHyD), where task difficulty is controlled by proof depth and a strict string-level scoring of the proposed hypothesis (*strong* correctness) gives a binary outcome to predict and intervene on. The statistical unit throughout is the task instance (row):

every effect is a pooled difference in $P(\text{strong})$ against an in-job baseline with row-cluster bootstrap confidence intervals, and every causal experiment was pre-registered with branch-complete decision rules before unblinding. Our answer, compressed to a sentence: *in a model that visibly predicts its own failures, the readable correctness signal and the causal repair channel are different objects — the entire readable subspace can be deleted without behavioral cost, while repair flows through an identified low-rank channel that carries the label-specific content of natural success yet is itself not decodably privileged relative to random subspaces of its rank — gauge and lever, anatomically separated.*

Concretely, we contribute: **(i)** a readable-stack erasure result with destructive matched-rank controls and a stated variance mechanism: nothing linearly readable about correctness is necessary (Section 5.2); **(ii)** a localized, compact causal channel — rank 8 at concept-mention tokens — that survives fresh-row and specificity guards (Sections 5.3–5.4); **(iii)** a necessity dissociation: the same natural class-mean vector is behaviorally null at full dimension and strongly reparative concentrated into the channel — and, in the removal direction, content-necessity at the channel's site that is broad rather than thin (Section 5.5); **(iv)** a label-specificity battery (shuffled labels, sign-flips, matched noise, random subspaces) that no steering paper in the verified corpus runs, falsifying both external reviewers' predictions; **(v)** causal control without decodable privilege — the inverse dissociation of the readable-but-inert readout; **(vi)** content-answer-free repair with a measured deployment profile: $+0.40$ where self-consistency scores 0.000, replicated on a disjoint fresh draw ($+0.28$), beneficial when misfired on already-correct rows at correct addressing — culminating in a *closed* answer-free repair loop: the gauge's scores over steered candidate states select the write-address (87–92% of the oracle effect — indistinguishable from oracle on draw 1, slightly below on draw 2; gold address on 43–54 of 57 rows) and the selected branch repairs $+0.24$ where matched-compute majority sampling sits at baseline (-0.02), replicated on a second disjoint draw ($+0.39$; Section 5.6); and **(vii)** a pre-registration methodology — decision rules committed before unblinding across 31 registered verdicts (branch-complete on confirmatory items), nine confirmed directional predictions, and token-level determinism gates — that we propose as a working standard for intervention claims in interpretability.

2 Dataset and Task

We generate InAbHyD abductive-reasoning instances: synthetic ontologies (taxonomic trees over invented concepts with inherited properties) in which the model must propose the hypothesis that best explains a set of observations. Tree height (1–4) controls proof depth; we run two task framings and use *property* (which property does the target concept have?) as primary, with *subtype* and a second model as scope checks. Each model answers 22,000 instances (1k/2k/3k/5k per height and task), scored two ways: *strong* correctness requires the exact gold hypothesis after canonicalization; *weak* accepts entailed variants. All causal work predicts and intervenes on strong correctness. Intervention experiments draw balanced height-3/4 rows from this pool under a strict provenance discipline: development rows (13), fresh-row guard sets (26+6), a 96-row natural-state capture set, and collateral sets of naturally-correct rows are pairwise disjoint, seeded, and frozen before use, so no basis, vector, or amplitude is ever fit on a row it is tested on (Appendix E).

3 Methods

Probes and the readable subspace. Correctness probes are class-balanced logistic regressions on residual-stream activations at the final prompt token, trained on held-out splits of the stage-1 dataset at five layers (L15–L53 in Gemma 3 27B). The *readable subspace* is constructed by iterative nullspace projection: at each layer we run INLP until probe AUC collapses to chance, orthonormalize the accumulated directions across all five layers, and obtain a rank-9 stack — everything any linear probe in our battery can read.

The erasure family. Erasure is a mean-projection clamp: at every token position, through prompt processing and decoding, the projection of the residual onto each stack direction is clamped to that direction's global scalar train-projection mean. Controls apply the *identical estimator and clamp* to matched-rank random stacks. Registered telemetry records within-forward-pass projection variance

Control	Mundane account it kills	Result
Random positions	any location works	null
Matched-norm noise	any perturbation this size helps	destructive
Random subspaces (norm/rank-matched)	any k directions work	null
Shuffled-label class-means	any difference-shaped vector works	null
Sign-flipped vector	geometry, not content	harmful
Shuffled-ridge coefficients	predicted content is generic	informative fail
Random stacks, identical estimator	erasure procedure is toothless	catastrophic
Dose ladder ($\times 1$, $\times 2$, executed arms)	more amplitude, more repair	monotone damage
Coordinate-permuted basis	spectrum-matched removal breaks	near-free
Typical-state (mean) replacement	the break is deleted energy	floors
Keep-only-basis complement deletion	the 8 dims carry the site	floors

Table 1: **Every positive result is bracketed by the control that would explain it away.** Sampling baselines (self-consistency, best-of-n) and collateral slices on naturally-correct rows complete the battery.

per direction (the mechanism check), and every arm reports $P(\text{strong} \mid \text{parsed})$ alongside $P(\text{strong})$ to separate content damage from format damage.

Interchange on the focus state. For a failing row, a *hint-delta* is the difference between residual states under a concept-hinted prompt and the unhinted prompt, computed on the longest common token block and restricted to the gold concept's mention positions. Repair arms add rank-k PCA reconstructions of these deltas at layer 30; bases are always fit leave-one-row-out (or on disjoint rows entirely), and additions are norm-matched per position to the reconstruction norms. Controls substitute random orthonormal subspaces, matched-norm Gaussian noise, random positions, or the delta mean alone at identical norms.

The class-mean family. Natural-state arms replace hints with observation: we capture unhinted concept-position states on 96 provenance-disjoint rows (balanced correct/incorrect by strong label), take per-row position-means, and form the class-mean $\bar{h}_{\text{correct}} - \bar{h}_{\text{incorrect}}$. Variants apply it raw or projected into the rank-8 basis; the *donor-free* variant fixes amplitude to a single pooled scalar so no per-instance information enters. Label controls recompute the mean under shuffled labels (identical geometry, norm, positions) or flip its sign; removal arms project states onto the orthogonal complement of a frozen basis, replace them with the dataset-typical mean state, or retain only the basis component.

Control taxonomy. Table 1 is the design's spine: each control is matched to the mundane account it eliminates.

Protocol. Every causal experiment was pre-registered with decision rules committed before unblinding; confirmatory items are branch-complete — a named branch for every reachable outcome pattern, wording included. One exploratory item (F(ii)) produced an outcome combination outside its enumerated branches; we flag it as such in Section 5.5, and its confirmatory battery (F(ii)-b/c) was registered and run afterwards. Nine registered directional predictions were confirmed. Amendment policy, stated as rule rather than practice: a registered result may be voided or amended only on evidence from an independent telemetry channel — a missing delivery fingerprint, an identically-zero degeneracy, a deviation audit — never on effect size or surprise; every amendment in the registry cites its channel. Generation is seed-deterministic per (row, arm, sample) — seeds are row-keyed ($s_0 + \text{row} \cdot 10007 + \text{arm} \cdot 101 + \text{sample}$, with lane variants adding layer or candidate terms; per-experiment s_0 and arm indices are pinned in the registration entries). Replication-gate arms reproduce prior jobs token-for-token ($416/416$, $736/736$, and $456/456$ generations across the necessity and addressing sequences alone); where a gate re-executes the identical (rows, seeds, arm) it is an integrity check on job-stitching rather than an independent replication, and we label the two cases. Statistics: row-cluster bootstrap (10k draws, fixed seed, percentile CIs) against in-job baselines; claim-bearing contrasts near zero additionally get BCa and leave-one-row-out checks; every null is reported with its minimum detectable effect (computed from the observed interval width — a post-hoc bound, not design power).

4 Related work

Readable but unsteerable. Our closest priors found the failure regime decodable (71.6%) with 29 fixed-linear-steering configurations null, concluding *entanglement* from a destructive LEACE erasure (arXiv:2605.05715, 2026), and detection-without-correction across seven model families (arXiv:2604.13068, 2026); arXiv:2605.23315 (2026) coins "epiphenomenal correctness" from cross-model convergence with small ablation effects. All three stop where our contribution starts: none localizes the causal variable or demonstrates repair, and our harmless full-stack erasure with destructive matched-rank controls adjudicates the entanglement account directly (Section 5.2). Correlational correctness probing supplies the readable half as background (arXiv:2604.05655, 2026; arXiv:2504.05419, 2025; arXiv:2602.06022, 2026).

Steering reasoning with difference vectors. The recipe family exists and we did not invent it: difference-in-means and related vectors can raise reasoning accuracy (arXiv:2601.19847, 2026; arXiv:2509.18116, 2025; arXiv:2505.12189, 2025), trained steering vectors recover most of RL's reasoning gains (arXiv:2509.06608, 2025), and probe-gated deployment is published (arXiv:2601.19847, 2026). None of this work shows necessity, label-specificity, or decodability dissociations; SAE-RSV (arXiv:2509.23799, 2025) shows raw vectors are mostly noise and filtering helps — we show a causally-identified filter is, in our setting, necessary, and what it retains is label-specific.

The methods bar. We adopt the state-encoding explicitness demanded by causal-abstraction critiques (arXiv:2507.08802, 2025) (fixed linear PCA bases, no trained alignment maps), the harmless-versus-pernicious vocabulary of arXiv:2511.04638 (2026) for the probe readout, mean-projection over INLP for the erasure operation (arXiv:2506.11673, 2025), and the subspace-illusion discipline of Makelov et al. (2023) (with Wu et al. (2024) as counterpoint) via fresh-row and basis-provenance guards. Steering-rigor critiques (arXiv:2501.17148, 2025; arXiv:2407.12404, 2024; arXiv:2410.17245, 2024; arXiv:2507.11771, 2025; arXiv:2411.07213, 2024) function here as the checklist our controls answer (Table 1).

Concurrent convergence. A concurrent frontier-lab result reports the same structural phenomenon on a different variable and model family: a small privileged subspace is causally dominant while decodable-but-inert content lives elsewhere (Anthropic Interpretability, 2026), independently replicated on a relative of our second model. We cite this as convergence on the anatomy — not prior art on correctness, repair, or the label battery.

5 Results

All effects are pooled differences in $P(\text{strong-correct})$ against in-job unhinted baselines, with row-cluster bootstrap 95% confidence intervals (10,000 draws, fixed seed); rows, not generations, are the unit of analysis, and unparsed generations score not-strong (so parse damage can only count against an intervention; arms with elevated parse failure additionally report $P(\text{strong} \mid \text{parsed})$). Every causal experiment below was pre-registered before unblinding (branch-complete on confirmatory items; the one registered-exploratory case is flagged where it appears), and replication-gate arms reproduce prior jobs token-for-token (Section 3).

5.1 The gauge: correctness is linearly readable

A linear probe on residual-stream activations at the final prompt token predicts whether the model's free-form answer will be strong-correct, before generation begins: test AUC rises from 0.786 at L15 to 0.902 at L53 in Gemma 3 27B, and reaches 0.940 in Qwen3.5-27B. The readout survives entity-name scrambling and beats every metadata baseline (height, prompt length, class balance). This much is consistent with a now-replicated literature: correctness and error are decodable pre-generation across model families, and the readable signal resists steering (arXiv:2605.05715, 2026; arXiv:2604.13068, 2026; arXiv:2605.23315, 2026). The question the rest of this section answers is what that readability is worth causally.

5.2 THE GAUGE IS NOT NECESSARY

Design. If the readable correctness signal participates in producing correct answers, deleting it should hurt. We deleted it completely. By iterative nullspace projection at each of the five readable layers — run to probe collapse, then orthonormalized across layers (Section 3) — we constructed the full rank-9 INLP-reachable correctness subspace — every direction any linear probe in our battery can read, at all five readable layers — and applied a mean-projection clamp at *every* token position, through both prompt processing and decoding. As controls, we applied the *identical* estimator and clamp to matched-rank random stacks. Sixteen balanced rows (8 naturally correct, 8 incorrect), six conditions, k=8 sampled generations each.

Result. Erasing the entire readable stack costs nothing detectable: $+0.047$ $[-0.070, +0.203]$ against the in-job baseline, with the single-axis continuity gate reproducing the historical null (-0.016 $[-0.094, +0.070]$). The identical operation on matched-rank random stacks is catastrophic: -0.380 $[-0.586, -0.182]$ pooled across three draws, with correctness among still-parseable outputs collapsing from 0.444 to ~ 0.01 — content demolition, not formatting damage. The paired contrast (readable stack − random family) is $+0.427$ $[+0.214, +0.643]$.

Mechanism, stated rather than hidden. The registered variance telemetry explains *why* the readable stack erases so cheaply: it carries 10–1300$\times$ less within-forward-pass projection variance than the random stacks — the clamp removes almost nothing where the readout lives. We therefore claim **non-necessity** of the readable subspace, never that the axis is "specially inert" among same-norm directions: the destructive controls prove the erasure family has behavioral teeth at matched rank and norm, and the low-variance mechanism explains the asymmetry. By the same discipline, the null is bounded, not proven zero: the minimum detectable effect at n=16 rows is ≈ 0.13. Two further honesty notes. *Application is verified, not assumed*: an erasure no-op would produce exactly this null, so we point to three independent checks — the destructive random-stack arms execute the identical estimator and code path with a different matrix (this verifies the shared machinery, though not the readable branch specifically: a branch-selective no-op would spare it, which is why the direct audit exists); the registered variance telemetry is read off the live hook during the erased forwards; and a direct application audit re-runs the clamp with a post-hook recorder, reporting projection deviations from the clamp targets under baseline and erased forwards, from which within-stack readability after the clamp is chance by construction once deviations vanish (Appendix A). The comparative claim is robust at this power regardless: the random-stack effect (-0.38) is roughly 3$\times$ the MDE, so readable $\ll$ random stands even where readable = zero is only bounded. *Predictability in retrospect*: given the variance telemetry, the null follows from the readable stack's low within-run variance — the experiment discriminates the load-bearing-readout account (and the rival entanglement account, which predicted damage) rather than surprising an informed reader who has already seen the telemetry; we state it as such.

What this adjudicates. The nearest rival account holds that readable correctness is *entangled* with task computation — their LEACE-style erasure damaged accuracy by 3.6pp, and they conclude the signal cannot be removed cleanly (arXiv:2605.05715, 2026). Our strictly larger erasure is free while matched-rank controls destroy, which excludes the destruction-on-removal form of entanglement in our setting. The weak, redundant-carrier form survives, and we say so: retrained probes still decode correctness after the scrub (information is redundant), yet no readable axis is load-bearing. Readability here is a gauge on the dashboard — which is precisely why months of steering it, by us (arXiv:2604.13068, 2026; arXiv:2605.23315, 2026, replicating), repaired nothing.

5.3 THE LEVER EXISTS AND IS LOCALIZED

The failure is a proposal failure. Behaviorally, the model's errors are not ignorance: given the gold hypothesis among candidates, accuracy jumps from 0.036 to 0.839; hinted with the target concept, to 0.955–1.000; yet the gold hypothesis is absent from sixteen sampled proposals on roughly two-thirds of failing rows. The model recognizes the answer it cannot propose — a *recognition gap* localized to target-concept focus.

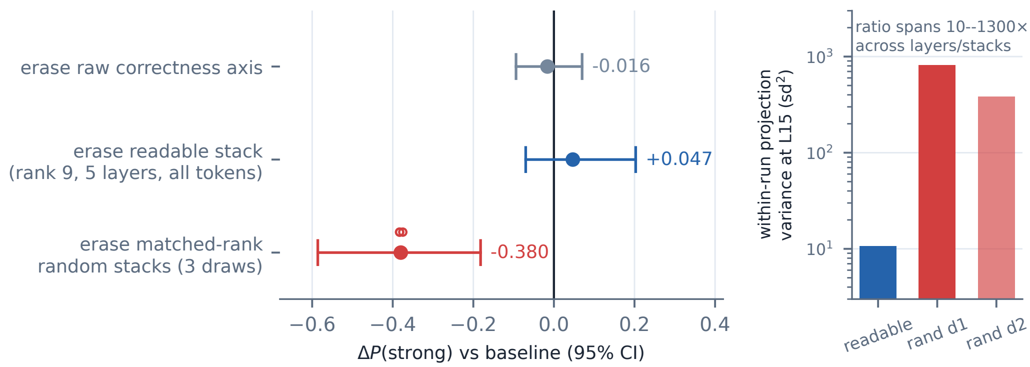

Figure 1: **Deleting everything readable costs nothing; deleting matched-rank random subspaces is catastrophic.** Pooled $\Delta P(\text{strong})$ with row-cluster bootstrap 95% CIs; 16 balanced rows, $k{=}8$. Inset: within-run projection variance per stack (log scale) — the readable stack carries 10–$1300\times$ less than the random stacks, the registered mechanism for cheap erasure.

Patching the focus state repairs generation. Replacing residual states at the gold concept's mention positions (layer 30) with their hint-conditioned counterparts repairs free-form generation on fresh rows disjoint from all development: $+0.255$ $[+0.111, +0.413]$ (the development-row anchor, reported as a labeled selected-row figure, is $+0.491$ $[+0.28, +0.71]$). Potency is positional: per-token, concept-mention patches are $\sim 6\times$ more effective than matched random positions, whose family is null. The token route is ruled out: masking the hint span costs nothing (natural decode attention to it is 0.5%), literal hint-token KV transplants are insufficient with attention verifiably flowing, and the masking$\times$reversion combination survives at 0.981 — the carrier is the unpatched layers' states at concept positions, not the hint tokens.

5.4 The lever is compact and specific

A rank-8 leave-one-row-out PCA basis of the hint-deltas carries the repair to rows that contributed to neither the basis nor the row selection: $+0.231$ $[+0.111, +0.365]$, i.e. 91% of the in-job patch effect (rank-4 under-transfers at 57% — rank 8 is the portable core). The pre-registered specificity ladder then separates the channel from its footprint: the rank-8 reconstruction repairs $+0.245$ $[+0.111, +0.394]$; its mean alone recovers 35%; random subspaces matched in mean, norm, rank, and position do nothing (-0.041 $[-0.154, +0.055]$); matched-norm noise at the same positions is destructive (-0.088 $[-0.180, -0.017]$); paired rank-8$-$noise is $+0.333$ $[+0.186, +0.489]$. The specific directions carry the repair. Geometrically, the core is disjoint from the gauge: its projection onto the readable stack is 0.0002, *below* the random-subspace null (0.0017), and while sparse-dictionary decoders see it, they see it redundantly — dictionary-visible, not sparse-small.

5.5 The necessity dissociation: what the channel carries

The lever of Sections 5.3–5.4 was built from hint-derived states. This section removes the hints, then the donors, then — in the final experiments — the assumption that the channel is where success itself lives. Throughout, the vector under test is the *natural class-mean*: the mean difference between states captured (without hints, at the same concept-mention positions, layer 30) on rows the model spontaneously gets right versus wrong — 96 capture rows, provenance-disjoint from every test row.

The same vector, five fates. At matched total norm on 26 held-out failing rows (this run was registered exploratory, and its observed outcome combination fell outside the enumerated branches — which is why the confirmatory battery below was then registered and run): (i) applied *full-dimensional*, the class-mean is undetermined: $+0.043$ $[-0.120, +0.207]$ (MDE 0.159 — a null is not licensed at this design); (ii) *concentrated into the rank-8 channel* by projection, it repairs strongly: $+0.341$ $[+0.202, +0.495]$ (BCa-stable; leave-one-row-out point range $[+0.315, +0.355]$; the licensed dissociation is the *paired* projected$-$raw contrast, $+0.298$ $[+0.130, +0.471]$); (iii) with *labels shuffled* before the mean is taken — identical geometry, norm, and positions — it does nothing: -0.043 $[-0.103, +0.006]$ pooled over four draws, paired real-vs-shuffled $+0.385$ $[+0.245, +0.534]$; (iv) with its *sign flipped*, it is actively harmful: -0.120 $[-0.245, -0.024]$; (v) made *fully donor-free* — direction from natural outcomes only, amplitude a single pooled scalar, zero per-instance answer information — it produces the largest point estimate in the F-series battery: $+0.399$ $[+0.245, +0.558]$ (statistically indistinguishable from the projected arm: paired $+0.058$ $[-0.005, +0.128]$). Both external reviewers predicted the shuffled arm would match the real vector; it did not. The labels carry the effect.

Against baselines nobody reports. On these hardest rows (baseline 0.120), majority-vote self-consistency (Wang et al., 2022) over eight samples scores 0.000 — errors are correlated across rollouts, so resampling regenerates the same wrong commitment — and a best-of-8 oracle ceiling reaches only 0.192 (both baselines are per-row selection accuracies on the same 26 rows; the steered per-row analogue is 16 improved, 10 unchanged, 0 worsened). Where prompting-side compute is the standard of comparison for steering claims (arXiv:2501.17148, 2025), this is the regime in which a causal write reaches states that sampling at matched compute did not.

The inversion. Is the channel privileged *readably*? A probe restricted to the rank-8 subspace decodes natural correct-vs-incorrect at 0.70–0.72 AUC — not distinguishable from matched random subspaces of the same rank (200-draw null: median 0.623, p95 0.721; full-dimensional ceiling 0.807) — i.e. the signal is diffuse enough that any 8 directions read well above chance, and the causal 8 read no better than their random peers. The identified repair subspace does not exceed the pre-registered random-subspace threshold for decoding natural outcomes. This is the mirror image of the field's decodable-but-not-causal refrain (arXiv:2605.05715, 2026; arXiv:2605.23315, 2026), demonstrated within one experiment pair: the readable thing is deletable (Section 5.2); the causal thing is not detectably more readable than its random peers. It also sharpens a live methods debate: raw difference-in-means vectors are mostly noise and filtering helps (arXiv:2509.23799, 2025) — here, causal filtering adds decisively (paired projected$-$raw $+0.298$ $[+0.130, +0.471]$; the raw arm alone is undetermined at MDE 0.159), and what the filter retains is label-specific content.

Deployment profile, measured. Firing the donor-free vector on rows the model already gets right *helps*: $+0.266$ $[+0.086, +0.469]$ ($0.727 \to 0.992$) — false-positive firings with correct addressing are beneficial, so gauge-gating is optional rather than load-bearing. The one answer-adjacent ingredient is *addressing*: written at all taxonomy-concept positions indiscriminately, the effect cancels ($+0.029$, n.s.), and every pre-named answer-free position policy fails. The intervention telemetry explains why: the vector acts as a positional commitment *command*, pulling proposals toward whichever concept is marked ($\sim$50% regardless of correctness) — content is the verb; the marked concept is the object. A registered follow-up replicates the repair on a second, fully disjoint draw of failing rows under its original in-job protocol ($+0.279$ $[+0.182, +0.377]$ on 57 rows; 62% of the $+0.447$ $[+0.293, +0.611]$ selected-row guard anchor — populations differ, so the ratio is descriptive), and a gold-only arm shows the *frozen* artifact also transfers — $+0.498$ $[+0.386, +0.607]$ on the same 57 rows at k=8, exceeding the in-job refit — so neither in-job fitting nor per-row amplitude is required at the gold address — indeed the frozen artifact wins the row-paired contrast outright ($+0.219$ $[+0.127, +0.316]$), which we read as estimator noise in the n≈26-donor leave-one-out refit rather than a freshness effect, and as evidence the vector's content is *stationary*: label-specific (shuffles null, sign-flips harm), stable across a month of the same generator (untested under distribution shift; and a donor-count ladder separating refit-estimator noise from genuine stationarity is registered), and behaviorally expressed as commitment at whatever address it is written to; wrong-address branches confirm the command reading — they shift proposals toward the fired concept ($+0.056$ $[+0.026, +0.086]$ over that concept's baseline rate) while leaving strong-correctness at baseline, since committing to a wrong concept cannot mint the gold hypothesis. The command lands $\sim 3\times$ harder where the prompt's evidence supports it (gold-address targeting lift $+0.18$ on both draws vs. $+0.056$ at wrong addresses) — the write interacts with prompt-consistency rather than acting as an address-agnostic push, a fact we file beside the open interference puzzle.

5.6 THE LOOP CLOSES

The program's final registered experiment composes its two working parts into a self-repair loop with no test-time answer information: given only the prompt, candidate concepts are enumerated (answer-free), the donor-free write fires at each candidate's positions in parallel branches (answer-free content), and the *gauge*'s score over each steered state selects the branch (answer-free selection). The composition passes its pre-registered primary: gauge-selected branches repair $+0.241$ $[+0.147, +0.342]$ over the in-job baseline and beat the matched-compute best-of-N majority comparator by $+0.259$ $[+0.167, +0.364]$; gauge selection could not be distinguished from oracle selection at our power (-0.022 $[-0.066, +0.018]$; the replication's contrast is -0.056 $[-0.121, -0.004]$, slightly below oracle), capturing 87–92% of the oracle effect; at the branch level the gauge's scores predict branch outcomes off-distribution at AUC $0.68/0.80$ across the two draws (a moderate ranker achieves near-oracle selection because top-1 regret depends on score separation at the top, not global AUC, and rows often contain several repairing branches; the spread across draws is variance, and the retained signal itself is a small new fact — a probe trained on natural states keeps usable, degraded signal on steered off-distribution ones, $0.90 \rightarrow 0.68$–0.80), picks the gold address on $43/57$ rows ($54/57$ under the milder frozen write), and beats random selection (by $+0.223$) and a self-ratification heuristic (by $+0.127$); non-gold branches sit at baseline (0.076 vs. gold 0.351), reproducing addressing specificity inside a single run. **The repair loop closes with no per-row answer information about the test problem at inference time**: candidates and addressing come from the prompt, and every supervised component — the gauge probe, the class-mean, the basis, the pinned amplitude — is fit on disjoint rows' correctness labels, never on the row being repaired. A protocol-identical fresh-draw replication (obligated by the registration) landed on a disjoint 58-row frame with every gate green: gauge-selected repair $+0.390$ $[+0.265, +0.515]$ — larger than the first draw — picking the gold address on $49/58$ rows and, on this draw, beating even the best-of-8 any-correct sampling ceiling (paired $+0.276$, CI > 0); the frozen-write arm replicates at $+0.513$ $[+0.399, +0.627]$. (The registered best-of-N comparator cannot be job-recorded for disjoint rows, so the replication uses in-job self-consistency@8, disclosed as an adaptation.) The closed-loop claim therefore stands on two disjoint draws. We also disclose a correction: the first two runs of this addressing family carried an execution defect — the branch *generations* ran without the steering hooks (only the gauge-scoring forward was steered), caught when the composition's first execution contradicted the in-job replication and confirmed by a telemetry audit (no positional-commitment signature in the branch outputs) — so those runs' behavioral outcomes, including a briefly-recorded frozen-write null, were voided and re-executed under a new registered hard gate: fired arms must show a targets-fired lift over baseline (delivery verified here: $+0.180$ $[+0.088, +0.276]$), or the run is execution-invalid by rule.

The necessity coda: what removal reveals. Sufficiency established, we ran the removal direction on 46 fresh naturally-correct rows (baseline 0.764; all gates verbatim against prior jobs, $416/416$ and $736/736$ generations token-identical). Projecting out the frozen rank-8 basis at the gold concept positions abolishes natural success (-0.666 $[-0.769, -0.557]$) while matched-rank random subspaces (-0.072) and a coordinate-permuted, spectrum-matched basis (-0.038) cost almost nothing. Registered telemetry, however, shows the fitted basis carries 97.7% of the state norm at those positions, so a second registered experiment matched the *energy* rather than the rank: replacing the states with the dataset-typical mean — preserving norm and the giant-magnitude dimensions, destroying row-specific content — floors success entirely (-0.764; zero surviving generations), as does removing the energy-optimal top-8 state-PCA subspace, and, decisively, as does *keeping only* the 8 repair dimensions while deleting their complement. A proportional 12% shrink at the same perturbation norm costs only -0.103, and the sign-flipped class-mean at matched norm beats matched-norm noise (-0.696 vs. -0.351; paired -0.345 $[-0.455, -0.243]$ — a seventh pre-registered directional prediction confirmed). A registered site-specificity control completes the coda: the same mean-replacement at matched count and energy costs a fraction at random positions (-0.236; row-paired $+0.527$ $[+0.397, +0.658]$ cheaper than the lever's site — a ninth confirmed directional prediction) and almost nothing at other concepts' mentions (-0.101; $+0.663$ cheaper) — the necessary content is not positions in general or concept mentions in general, but the lever's address. The joint reading is scoped and, we believe, more interesting than the thin-channel necessity it replaces: at the lever's site, natural success requires the state's specific content *wholesale* — **the lever is write-compact, but what it writes into is carried broadly**. A reviewer-prompted overlap analysis sharpens the energy accounting: the basis's first two components are nearly axis-aligned with the model's outlier dimensions (participation ratios 2 and 4; the top-8 outlier coordinates hold 98% of state energy,

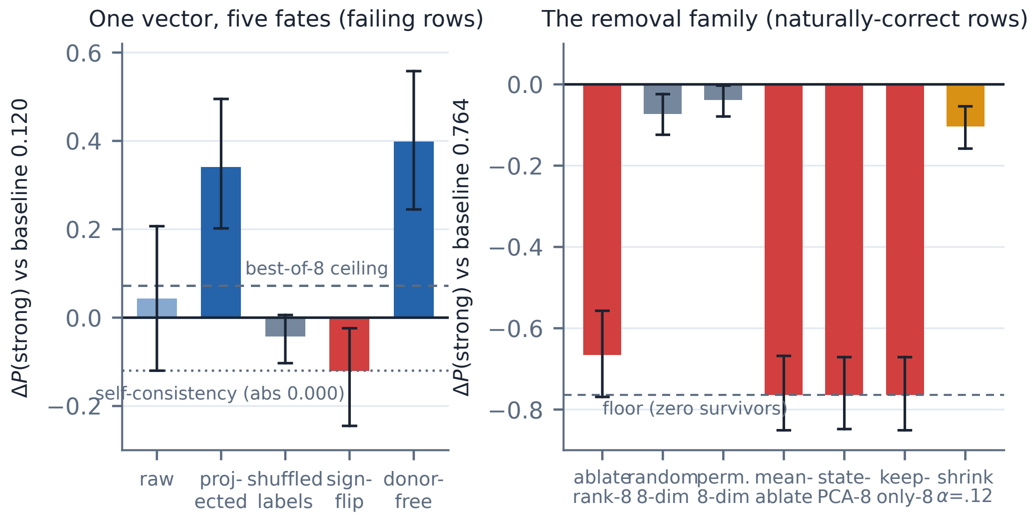

Figure 2: **One vector, five fates — and the removal side.** Left: the natural class-mean at matched norm is null full-dimensional, potent concentrated into the channel, inert with shuffled labels, harmful sign-flipped, and strongest fully donor-free ($+0.399$ on a 0.120 baseline; self-consistency 0.000, best-of-8 0.192). Right: on naturally-correct rows, every energy-preserving content destruction at the lever's site floors success (the rank-8 projection, which removes 97.7% of the norm, nearly floors at -0.666), while content-preserving perturbations are an order cheaper. All removal arms pre-registered confirmatory; left-panel arms (i)–(ii) are from the registered-exploratory F(ii) run whose confirmatory battery follows them; row-cluster bootstrap 95% CIs.

and the basis captures 95.5% against their 97.8%), so the removal-side results are stated as energy-bound — which the controls already enforce (the energy-optimal state-PCA-8 removal also floors; the coordinate-permuted spectrum-matched basis, whose mass lands off the outlier coordinates, is near-free). Write-side specificity is not outlier-explained: shuffled-label vectors share the outlier profile exactly and repair nothing.

5.7 Scope and cross-model replication

Positive lever claims are Gemma-3-27B, property task. The scope checks: **Subtype** replicates the behavioral story (recognition gap $33/48$; candidates 1.000) but residual-state repair is only suggestive — a weaker, row-sparse analogue at a different depth ($+0.117$ $[+0.008, +0.254]$ at L35, carried by $3/13$ rows; L35 was selected by a three-layer causal screen (L35/L50/L53) and this is its targeted two-seed replication, so the discovery step carries multiplicity the replication does not erase). **Qwen3.5-27B** replicates raw-axis non-necessity ($+0.070$ $[-0.031, +0.188]$; its controls are also non-destructive, so this is support, not a replication of the destructive-control profile) — and then the port becomes informative in its own right. The same recognition-gap failure mode exists (hint lift $+0.523$); a concept-position carrier exists at relative depth 0.67 vs. Gemma's 0.48 (concept-replace $+0.175$ $[+0.100, +0.258]$); the channel rank is $\sim$16 vs. 8, with recovery climbing $48\% \rightarrow 95\%$ of the carrier across ranks $8 \rightarrow 64$ while random 64-dim subspaces at matched norm sit at baseline. The content result *inverts* in the projection direction: in Qwen the *raw* class-mean repairs ($+0.120$ $[+0.016, +0.224]$; real-vs-shuffled paired $+0.182$ $[+0.070, +0.299]$; sign-flip harmful, a fifth confirmed registered prediction) and projection into the compact channel *dilutes* it, where Gemma's projection added ($+0.298$ paired). Gemma's raw arm, undetermined at MDE 0.159, cannot exclude a Qwen-sized raw effect — the licensed inversion is in the projection contrasts, and we state it that way. Meanwhile no outcome information was detectably decodable at the channel's address (AUC $0.504/0.483$ across two draws, SE $\approx$0.06 at n=96; vs. Gemma's 0.807). Instead, Qwen's outcome information is weakly decodable at the *final prompt token* (0.66–0.81 across two independent 96-row draws), and writing the class-mean at that readable site does not repair (label-shuffled matches real exactly, paired $+0.005$; a sixth confirmed prediction). The anatomy is conserved, its coordinates

	Gemma / property	Gemma / subtype	Qwen / property
Gauge readable	✓ 0.90	✓	✓ 0.94
Readout-axis non-necessity	✓	✓	✓ (controls also null
Full-stack non-necessity	✓	—	—
Lever repair + specificity	✓ rank 8	suggestive	✓ motif, $k^*\approx 16$
F-series dissociations	✓	—	✓ inverted (raw-poten
Content necessity at lever site	✓ broad, not thin	—	—
Closed answer-free loop (gauge-select × write)	✓ +0.241 / +0.390 (two disjoint draws)	—	—

Table 2: **Scope matrix.** Checks are pre-registered pooled results; "—" means untested, not negative.

are not: **Gemma separates gauge from lever by subspace at one position; Qwen separates them by position at one layer. Gemma concentrates, Qwen distributes — the recipe is cross-model, the compression is not.** A calibration-stage observation promotes a further caveat: the two models' *failure modes* differ in kind. Failing Gemma rows usually never name the gold concept (baseline targeting $\sim$0.4) — a commitment failure; failing Qwen rows name it $\sim$0.9 of the time and still build the wrong hypothesis — a construction failure downstream of concept selection. Whatever the write repairs in Qwen, it is not concept-selection, so the cross-model story should be read as "one intervention, possibly two mechanisms" until a hypothesis-level delivery diagnostic (registered for the port's analysis) adjudicates.

6 Discussion and limitations

Location is not control, in either direction. The program's two dissociations bound what probe evidence can license. For monitoring: a high-AUC correctness probe can be a perfectly good gauge while reading a subspace the computation never consults — deletable at zero cost. For steering: the potent content can live where no probe finds it, so steering nulls on readable directions are uninformative about controllability. The lineage's raw difference-in-means vectors plausibly pay a dilution tax and a damage ceiling that causal filtering removes — a testable prediction (in Gemma, projection added +0.298 paired; in Qwen, where the state distributes, raw suffices and projection dilutes), which the concentrated-vs-distributed contrast turns into a measurable, model-level design property.

Endogeneity, honestly. The channel repairs without being decodably privileged, so we claim *exogenous mediation*: the vector carries content *about* natural success without being shown to be the model's own per-row decision variable (as far as linear tools see). The removal experiments sharpen this: at the lever's site, natural success requires the state's specific content wholesale — replacing it with a typical state at preserved energy abolishes success, as does keeping only the 8 repair dimensions — so the channel is a write-compact port into a broadly-carried state, not a thin wire that computation runs through. We say "an identified sufficient channel," not the mechanism; sufficiency was searched for, and pre-registration is the mitigation we offer, not a proof of uniqueness.

The gauge is not the verbal channel — and not useless. Asked whether its own replayed answer was "exactly correct" (a 2,000-row census on Gemma 3 27B), the model verbalizes "no" near-uniformly: on 85% of answers it in fact got right, with 97.5% unanimity across samples and a confident-wrong cell of 0.9% — while its linear gauge reads upcoming correctness at AUC 0.90 on the same task. The verbal surface is also extremely elicitation-fragile: an identical-rows census that drops the word "exactly" flips yes-rates from 8.2% to 50.2% and becomes informative (68% yes on correct vs. 32% on wrong) — so the knowing-saying gap is protocol-dependent at the verbal surface, while the internal readout is not. Either way, whatever the probe reads, it is not the model's verbal self-report channel — and a registered factorization makes this causal: on a balanced 2×2 frame of (objective correctness × verbalized self-judgement) with both signals linearly readable in the same states (SJ at AUC 0.81), the gauge sides with *objective* correctness on 84% of conflict rows (diff -0.69 $[-0.84, -0.50]$), and the lever repairs confident-wrong rows as strongly as ordinary failures (+0.62 vs. +0.50; difference CI spans zero, MDE 0.23 — an eighth confirmed registered prediction). Gauge and lever both live on the objective-correctness side; the self-report channel is causally bypassed by both. And the gauge, causally inert as a lever, has a real causal *job* in the one role its nature permits: as the *selector* in the closed answer-free repair loop, capturing 87–92% of the oracle effect (+0.241 and +0.390 repair through its choices on two disjoint draws; Section 5.6). Gauge and lever remain different objects; the gauge's job is reading — and it reads steered states well enough to do causal work there, though degraded ($0.90 \to 0.68$–0.80).

Open anomalies and limits. Misdirection asymmetry is partly an estimand artifact — wrong-address writes do shift proposal distributions toward their targets (+0.056, CI $>$ 0) while ΔP(strong) cannot register success for a wrong target by construction — but the residual anomaly stands: no tested write produces specific wrong *strong-scored* answers, and the attractor-strength account has the wrong sign. The all-positions cancellation (+0.03) beside wrong-address-neutral and gold-address-positive implies interference between simultaneous writes that naive superposition does not predict; open. Positional addressing is the one answer-adjacent ingredient of the answer-free repair (every pre-named answer-free position policy fails). Quantitative claims rest on n=13–57 row sets with MDEs of 0.10–0.16 on nulls; positive results are one task family, one model (with the stated Qwen and subtype scope); brittleness and entropy telemetry are unrun. The closed loop is an existence proof, not a method: it closes because candidate addresses are enumerable from the prompt, a property most tasks lack. Rows are independently generated ontologies (no shared templates beyond the generation grammar), so row-cluster bootstrap is the natural unit. Future work, in registered order: the layer×position information map in Qwen; the closed loop's port to the second model; and porting the necessity coda to the second model.

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A FROZEN ARTIFACTS, SEEDS, AND DETERMINISM GATES

Every registered arm consumes only artifacts frozen before unblinding. Table 3 lists them. Interventions are deterministic by construction: sampling seeds are row-keyed per sample ($\mathrm{seed} = s_0 + \mathrm{row} \cdot 10007 + \mathrm{arm} \cdot 101 + \mathrm{sample}$; lane variants add layer or candidate-index terms, and each experiment's s_0 and arm indices are pinned in its registration entry), so shard layout cannot change any generation, and every follow-up run re-executes its predecessor's baseline arms and requires token-level verbatim equality before its new arms are read (gate counts across the program: $416/416$, $736/736$, and $456/456$ verbatim replays across the K/K$'$ and L/L$'$ sequences; earlier series gated identically). Registered lanes are pinned to one venue (bf16 on 2×A40) because bf16 arithmetic is bit-identical across GPUs of one architecture but summation order differs across architectures, which breaks verbatim gates. Statistical claims use row-cluster bootstrap (10,000 draws, seed 20260704, percentile intervals); every null carries a minimum detectable effect computed from its own interval width (a post-hoc bound, not design power).

artifact	frozen from	role
class-mean vector	96-row capture (job 458416)	repair content source
rank-8 basis	states `npz`, 26 `concept_delta` keys (job 458431)	write subspace
pinned write norm	3708.2628 (recomputed, hard-fail on drift)	energy control
gauge direction	row 0 of `L53_inlp_stack` (INLP stack `npz`)	readable axis
donor row set	16 correct rows, F(ii)-c registration	LOO donor pool
row selections	seeded draws (e.g. 20260812 for the L-series)	sampling frames

Table 3: Frozen artifacts consumed by registered arms. Each is stored with its producing job id; registrations name them before any outcome is observed.

B NEGATIVE-RESULTS CATALOG

Pre-registration makes negative results reportable rather than discardable. Table 4 catalogs every registered null or failed branch, with its resolution where a follow-up resolved it. None were dropped silently; each fired a pre-named branch of its registration.

finding (registered null / failure)	effect [CI] or MDE	resolution / status
erasing the raw readout axis: no cost	-0.016 $[-0.094, +0.070]$	thesis-supporting null
erasing the full readable stack: no cost	$+0.047$ $[-0.070, +0.203]$	thesis-supporting null
Qwen rank-8 transplant misses	$+0.083$, MDE 0.12	G3′: channel rank k^*=16
Qwen projected-16 content transfer null	—	G6′: full-state raw transfer passes (proj-64 flagged, closed positive)
writing at Qwen's readable site (final token)	paired $+0.005$; shuffled = real	6th confirmed prediction: position dissociation
frozen-basis write on fresh rows (first L run)	void, not null	execution defect (unhooked generation); corrected gold-only arm: FROZEN-TRANSFERS $+0.498$
misdirection toward specific wrong answers	attractor account: wrong sign	open anomaly
answer-free position policies (all pre-named)	all fail	positional addressing remains answer-adjacent
endogeneity (channel as the model's own decision variable)	not shown at this layer/rank	claim limited to exogenous mediation
mean-component-only repair	$+0.087$ $[+0.019, +0.173]$	partial; full recipe needed

Table 4: Registered negatives. MDEs state the effect size the sample could have detected; "null" therefore means "smaller than MDE," never "exactly zero."

C PROVENANCE MAP

Every number in the paper traces to a pooled verdict document in the repository, which names its jobs, gates, and registration item. Table 5 maps the paper's claims to that chain.

claim (section)	item	verdict document (jobs)
gauge readable (§5.1)	—	stage-1 probe suite
erasure at zero cost (§5.2)	C/D/E	subspace-erasure, control-matching, and readable-stack summaries (456912–457210; 458403–458414)
lever localization, rank, specificity (§5.3–5.4)	A/C/E + guard v2	rank-k guard v2 pooled (458374/458375); rank-8 specificity pooled (458401/458402)
content battery (§5.5)	F(i)/F(ii)/F(ii)-b/c, H	F-series pooled summaries; position-policy summary
necessity coda (§5.5)	K/K′	necessity pooled (459836–459838); prime pooled (459847/459849/459850)
Qwen port (§5.7)	G0–G6′, J1/J2	`qwen_*` summaries (458461–458525)
closed loop: selector, fresh-draw repair, frozen transfer (§5.5)	L/L′/L″	self-address pooled (461624–461633); prime pooled (461648/461649); loo pooled + rider (461886–461890)
self-judgement census (§6)	recon	census summary (H100 batch jobs)

Table 5: Claim-to-artifact provenance. The registration registry (`causal_handle_directions.md`) is version-controlled; each item's decision rules were committed before its jobs ran.

D PARSING AND SCORING DIAGNOSTICS

All arms are scored on $\Delta P(\text{strong})$ against the in-job unhinted baseline, with unparsed generations scored not-strong (so parse failure can only hurt, never help, an intervention). Registered parse gates for claim-bearing repair arms: $< 5\%$ parse failure passes; 5–20% is flagged and additionally reported as $P(\text{strong} \mid \text{parsed})$; $> 20\%$ voids the arm. No claim-bearing repair arm exceeded the void bound; flagged arms exist (e.g. up to 11.7% on K′ removal arms, 18.3% on the exploratory F(ii) projected arm) and in every case $P(\text{strong} \mid \text{parsed})$ preserves the reported ordering. *Destructive* controls are a different regime by design: breaking the model breaks its formatting too, so their parse rates run as low as 0.66 (random-stack erasure draws), and their claims rest jointly on $\Delta P(\text{strong})$ and on $P(\text{strong} \mid \text{parsed})$ collapsing (0.008–0.017 for the random stacks) — content demolition confirmed among the outputs that still parse, not an artifact of format damage.

E REGISTRATION EXCERPTS

Registrations are appended to the registry before any outcome is observed, and are branch-complete: every observable outcome maps to a pre-named verdict sentence. As an exemplar, the composition item (L″, passed; replicated on a disjoint fresh draw) registers: PRIMARY PASS iff the gauge-selected branch's pooled $\Delta P(\text{strong})$ CI excludes zero AND the paired contrast against the recorded best-of-N majority-vote comparator excludes zero; pre-named failure branches include selector-write interference (selector ranks gold correctly but the write undoes it) and gold-instability (the gold address is not stable across candidate draws); a PASS obligates a fresh-draw replication (L‴) before any claim upgrade. After the execution-defect correction (Section 5.5), the rerun adds a hard delivery gate: fired arms must show a targets-fired lift over baseline (CI > 0) or the run is execution-invalid by rule — a steering-signature positive control we now consider mandatory for any intervention lane. The full registry, with per-item commit hashes, ships in the artifact bundle.

Artifact bundle. The bundle is the project repository: the registration registry, every pooled verdict document cited in Appendix C, the frozen `npz` artifacts of Table 3, and the task-facing surfaces a reconstruction needs — the system and user prompt templates (`src/config.py`), the hint-prompt constructors, the strong-correctness canonicalization rules (`src/gemma3_parse.py` and the scoring module), and the probe training code for the readable stack. Per-experiment s_0 values, arm index maps, and INLP stopping criteria live in the registration entries and run manifests alongside the code that consumed them.